Home Network Design & Security

Project Overview

A home network simulated using Cisco Packet Tracer, focusing on VLAN segmentation, inter-VLAN routing, DHCP, DNS, web hosting, guest Wi-Fi, and network access control.

Network Design

The network is divided into three VLANs:

- **MAIN (VLAN 10):** Personal computers, server, and printer.
- **IoT (VLAN 20):** Dedicated network segment for IoT devices.
- **GUEST (VLAN 30):** Guest wireless devices with restricted access to internal networks.

Devices & Technologies

Device	Network Role	Function
Cisco 2811 Router	Router	Connects different VLANs, gives devices IP addresses, and controls guest network access.
Cisco 2960 Switch	Switch	Connects wired devices and separates them into different VLANs.
Server-PT	Server	Stores the local website and helps devices find it using DNS.
WRT300N	Wireless access point	Provides Wi-Fi for guest devices.
Main PCs	Main network devices	Connect to the main network and access the local website.
Guest laptop	Guest network device	Connects to the guest Wi-Fi and tests access restrictions.
IoT laptop	IoT network device	Represents an IoT device and tests the IoT network connection.
Printer-PT	Network printer	Represents a printer that can be connected to the network.

IP Addressing

VLAN	Network	Gateway
MAIN — 10	192.168.10.0/24	192.168.10.1
IoT — 20	192.168.20.0/24	192.168.20.1
GUEST — 30	192.168.30.0/24	192.168.30.1

Key Features

- VLAN segmentation and 802.1Q trunking.
- Router-on-a-stick inter-VLAN routing.
- DHCP for automatic IP assignment.
- DNS resolution using home.lab.
- Local website hosted on Server-PT (192.168.10.10).
- Guest Wi-Fi configuration.
- Guest network isolation using an extended ACL.

Security

The GUEST-ISOLATION ACL restricts guest access to the MAIN and IoT subnets while permitting DHCP requests and DNS queries to the designated server.

IoT isolation from the MAIN network has not yet been implemented.

Testing Result

No.	Test	Expected Result	Status
1	MAIN PC DHCP	Receives an IP address in VLAN 10	Passed
2	IoT DHCP	Receives an IP address in VLAN 20	Passed
3	Guest DHCP	Receives an IP address in VLAN 30	Passed
4	MAIN gateway connectivity	Can ping 192.168.10.1	Passed
5	IoT gateway connectivity	Can ping 192.168.20.1	Passed
6	Guest gateway connectivity	Can ping 192.168.30.1	Passed
7	Guest DNS resolution	home.lab resolves to 192.168.10.10	Passed
8	Guest access to MAIN server	Direct access is blocked by the ACL	Passed
9	MAIN website access	MAIN PC opens http://home.lab	Passed
10	IoT access to MAIN server	Server responds before IoT isolation is configured	Passed

Tools

Cisco Packet Tracer · Cisco IOS CLI · IPv4 · VLANs · DHCP · DNS · Extended ACLs

Conclusion

This project demonstrates the design and configuration of a segmented home network, providing practical experience in routing, switching, network services, and basic network security.